

PENINAH WANJIKU MBURU

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PROFILE

Final-year Mathematics and Computer Science student at JKUAT (graduating 2026), with a focus on data analytics, machine learning, and software engineering. I build tools that are explainable, fair, and grounded in real-world needs — drawing on hands-on experience at M-Pesa Africa, Convergent Networks, and Bobplus Africa.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Sept 2022 — Present

Bachelor of Science in Mathematics and Computer Science

Relevant coursework: Artificial Intelligence · Mobile Computing · Cloud Computing · Time Series Analysis · Simulation & Modelling · Partial Differential Equations II · Multimedia Systems

Moi Girls' School Nairobi

Sept 2019 — Apr 2022

Kenya Certificate of Secondary Education (KCSE) — B+, 72 points

TECHNICAL SKILLS

Languages: Python · JavaScript · Dart · SQL · HTML/CSS · C

AI / ML: Scikit-learn · Random Forest · SHAP · K-Means · Naive Bayes · Pandas · NumPy

Mobile & Backend: Flutter · Firebase · Node.js · Express · WebSockets · Cloud Functions

Tools & Platforms: AWS · Git/GitHub · Streamlit · Power BI · Gephi · Nodemailer

Mathematics: Statistics · Time Series Analysis · PDEs · Simulation & Modelling

WORK EXPERIENCE

Bobplus Africa

Jun 2025 — Jan 2026

Customer Success Executive Intern

- Served as the first point of contact for merchant support queries via Microsoft Teams and email.
- Monitored merchant accounts for recurring transaction failures and flagged patterns for investigation.
- Documented common merchant issues and contributed to the internal knowledge base for the CS team.
- Collaborated with technical and operations teams to resolve merchant-facing platform issues.

M-Pesa Africa (Safaricom PLC)

Aug 2024 — Oct 2024

Apprentice

- Collaborated in the creation of a lookup table on AWS.
- Assisted in data modelling and SQL database optimisation.
- Worked with data scientists to align analysis results with business goals.
- Analysed large datasets using statistical methods.
- Gained proficiency in AWS, SQL, and Gephi.

Convergent Networks

May 2023 — Aug 2023

Intern

- Assisted in network troubleshooting and creating network diagrams and communications.
- Worked with the IT team to streamline internal data processes.
- Troubleshoot hardware and software problems and performed routine network maintenance.

PROJECTS

**BobBot — CS Automation System | Node.js · Express · WebSockets ·
Microsoft Bot Framework · Nodemailer · Claude API**

2026

- Built a full customer support automation system for Bobplus Africa based on firsthand CS experience.
- Handles merchant queries on Microsoft Teams with a transaction monitor that escalates alerts at 12, 24, and 48 hours.
- Real-time CS dashboard gives agents a unified view of merchant chats, alerts, and transaction history.

**AttendX v2 — Smart Attendance System | Flutter · Dart · Firebase ·
GPS Geofencing · Bluetooth Beacons · Cloud Functions**

2026

- Final year project: automates student attendance at JKUAT using GPS and Bluetooth beacon verification.
- Three user roles (student, teacher, admin); anti-proxy via device binding and biometric authentication.
- Cloud Functions auto-close sessions and push at-risk alerts when attendance falls below 75%.

**Credit Scoring via Mobile Money Behaviour | Python · Random Forest
· SHAP · K-Means · Streamlit · Scikit-learn**

2026

- Group project: 7-phase ML pipeline simulating 1.96M synthetic M-Pesa transactions for 5,000 users.
- Random Forest achieved AUC-ROC of 0.9958, rescuing 97 creditworthy borrowers from wrongful loan rejection.
- SHAP explainability and Streamlit dashboard with live What-If simulator. Supervised by Dr. P.R. Kiogora, JKUAT.

**Titanic Survival Prediction | Python · Pandas · Scikit-learn · Seaborn ·
Matplotlib**

2026

- Full ML workflow: EDA, cleaning, feature engineering, and evaluation of Decision Tree, Naive Bayes, and Random Forest.
- Random Forest achieved 82.1% accuracy; sex (28%) and fare (27%) identified as strongest survival predictors.

REFERENCES

George Gakungu — Director, Convergent Networks

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Asumpta Muia — Senior Executive Assistant, M-Pesa Africa (Safaricom PLC)

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